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Code No: N0523 /R07

Set No. 1

IV B.Tech. I Semester Regular Examinations, November 2010
MOBILE COMPUTING
(Common to Computer Science & Engineering, Information Technology and Electronics & Computer Engineering)

Time: 3 Hours Max Marks: 80

Answer any Five Questions
All Questions carry equal marks
****

1. (a) Explain the architecture of mobile computing.
   (b) Where and when can collisions occur while accessing the GSM system? Compare possible collisions caused by data transmission in standard GSM, HSCSD and GPRS.

2. (a) What are the benefits of reservation schemas? How are collisions avoided during data transmission, why is the probability of collisions lower compared to classical Aloha?
   (b) What is CDMA? Explain in detail.

3. (a) What is the main purpose of registration of a mobile node? Explain in detail.
   (b) Write short notes on IPv6.

4. (a) Describe transaction oriented TCP.
   (b) Explain Mobile TCP. How does a supervisory host send TCP packets to the mobile node and to a fixed TCP connection.

5. (a) Explain about power aware computing.
   (b) Explain the data recovery processing in detail.

6. (a) Explain push based data delivery mechanisms in detail.
   (b) Explain the following selective tuning and indexing techniques.
      i) Directory method ii) Flexible indexing method

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7. (a) How does dynamic source routing handle routing? What is the motivation behind dynamic source routing compared to other routing algorithms for fixed networks.
   (b) Describe security problems in MANETs.

8. (a) Explain the functions of radio, baseband and link manager in Bluetooth.
   (b) Explain about the wireless transaction protocol of the transaction layer.

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Set No. 2

IV B.Tech. I Semester Regular Examinations, November 2010
MOBILE COMPUTING
(Common to Computer Science & Engineering, Information Technology and Electronics & Computer Engineering)

Time: 3 Hours Max Marks: 80

Answer any Five Questions
All Questions carry equal marks
****

1. (a) Explain the applications of mobile computing.
   (b) Name the main elements of the GSM system architecture and describe their functions. What are the advantages of specifying not only the radio interface but also all internal interfaces of the GSM system.

2. (a) Define the problem of Hidden and Exposed terminals. What happens in the case of such terminals if Aloha, Slotted Aloha, reservation Aloha or MACA is used?
   (b) Explain TDMA and its features.

3. (a) What are general problems of mobile IP regarding security and support of quantity of service?
   (b) What is meant by Encapsulation? Explain various types of Encapsulation.

4. (a) Write brief notes on congestion control in traditional TCP.
   (b) Compare several enhancements to TCP for mobility.

5. (a) Explain the concept of cache invalidation mechanisms.
   (b) Explain the following concepts in mobile Environments
   (i) Data cache maintenance (ii) Web cache maintenance.

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Code No: N0523 /R07

Set No. 2

6. (a) Explain the Hash based and Index based selective tuning and indexing techniques.
   (b) Write about communication asymmetry with the help of a diagram.

7. (a) Compare the reactive and proactive routing protocols.
   (b) Explain the properties of MANETs.

8. (a) What are advantages and problems of forwarding mechanism in Bluetooth networks regarding security ,power Saving and network Stability?
   (b) Explain about the wireless application Environment (WAE) logical model.

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Code No: N0523 /R07 **Set No. 3**

IV B.Tech. I Semester Regular Examinations, November 2010
MOBILE COMPUTING
(Common to Computer Science & Engineering, Information Technology and Electronics & Computer Engineering)

Time: 3 Hours Max Marks: 80

Answer any Five Questions
All Questions carry equal marks
****

1. (a) What is mobile computing ?
   (b) What are the drawbacks of mobile and wireless devices.
   (c) Explain the teleservices of GSM.

2. (a) What is the motivation for a specialized MAC.
   (b) Who performs the MAC algorithm for SDMA? What could be possible roles of mobile stations, base stations and planning from the network provider?
   (c) Define the Hidden and Exposed terminals problem.

3. (a) What is IP packet delivery? Explain it.
   (b) What is Encapsulation? Explain minimal Encapsulation.

4. (a) Compare the different types of transmission errors that can occur in wireless and wired networks. What additional role does mobility play?
   (b) Explain mobile TCP and write the advantages and disadvantages of mobile TCP.

5. (a) Explain the stateless asynchronous and stateful synchronous cache invalidation mechanisms in detail.
   (b) Write the advantages and disadvantages of stateless synchronous and stateful asynchronous cache invalidation mechanism.

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Set No. 3

6. (a) Explain the Hybrid data delivery mechanism.
   (b) Explain about communication asymmetry .

7. (a) What are the benefits of location information for routing in ad-hoc networks, which problems arise?
   (b) Explain destination sequence distance vector routing algorithm in MANETs.

8. (a) Explain the physical and networking layers of WAP.
   (b) Explain the packet format at the base band layer of Bluetooth.

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Set No. 4

IV B.Tech. I Semester Regular Examinations, November 2010
MOBILE COMPUTING
(Common to Computer Science & Engineering, Information Technology and Electronics & Computer Engineering)

Time: 3 Hours Max Marks: 80

Answer any Five Questions
All Questions carry equal marks
****

1. (a) Explain the system architecture of GSM.
   (b) Why is a new infrastructure needed for GPRS, but not for HSCSD ? which components are new and what is their purpose.

2. (a) What is reason for failure of MAC schemes in wireless networks. What is done in wired networks to avoid this affect.
   (b) Explain the term interference in the space, time, frequency and code domain. What are counter measures in SDMA, FDMA and CDMA systems.

3. (a) Explain client initialization via DHCP.
   (b)Explain packet flow if two mobile nodes communicate and both are in foreign networks. What additional routes do packets take if reverse tunneling is required.

4. (a) What is the reaction of standard TCP in case of packet loss. In what situation does this reaction make sense and why is it quite often problematic in the case of wireless networks and mobility.
   (b) Explain snooping TCP. What are its advantages and disadvantages.

5. (a) Explain the data cache maintenance and web cache maintenance in mobile environments.
   (b) Explain the issues relating to QoS in mobile networks.

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Set No. 4

6. (a) Explain pull based data delivery mechanisms in detail.
   (b) Explain the hash-based and distributive index based selective tuning and indexing techniques.

7. (a) What are the differences between AODV and standard distance rector rooting.
   (b) Write the applications of MANETs.

8. (a) Describe J2ME features.
   (b) Explain about the wireless data gram protocol of transport layer.

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